Week 3 Mandatory Hand-in: Consumption

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Purpose. This assignment is based mainly on the consumption lecture, with a short link to the following lecture's aggregate-demand mechanism. It asks you to move from equations to computation, figures and economic interpretation. The core question is: when does consumption respond to current income and cash flow, and when does it respond mainly to lifetime resources?

Exam qualification. This is a mandatory pass/fail hand-in. You must pass the assignment to qualify for the exam. You may work in groups of at most 4 students. Use the exercise class to work on the assignment and to ask questions about modelling choices, coding and interpretation.

Files.

- Student starter notebook: `week03_consumption_student_start.ipynb`
- Data file: `week03_fred_consumption.csv`
- Submit: your completed notebook, renamed `week03_consumption_GROUPNAME.ipynb`

What counts as a pass? A passing submission must:

1. run from a clean kernel without errors;
2. contain the requested figures and tables with clear labels;
3. include short Markdown interpretations after the main outputs;
4. correctly use the intertemporal budget constraint, the Keynes-Ramsey logic, the PIH annuity factor and the borrowing-constraint mechanism;
5. include a group-specific household profile in Part 3, so that the hand-in is not just copied from the starter notebook;
6. end with a concise economic narrative that refers to your own numbers.

Advice. Do not try to write a long essay. A good answer is precise: it explains what was computed, what the result means, and which consumption mechanism is responsible. Several subtasks ask you to refer to specific numbers from your own output; the final narrative will be assessed on whether you do this.

Part 1. Current income and the Keynesian benchmark

Use

$$C = \bar{C} + bY^d, \qquad \bar{C} = 200, \qquad b = 0.75.$$

1. Plot consumption against disposable income for $Y^d \in [100, 1200]$.
2. Compute MPC and APC at $Y^d \in \{400, 600, 800, 1000\}$ and display the results in a table.
3. Explain why the prediction for APC is useful in the cross-section but problematic for long-run aggregate data.

Part 2. Two-period consumption and the real interest rate

Use the two-period model from the lecture:

$$H_1 = Y_1^d + \frac{Y_2^d}{1+r}, \qquad C_1 = \theta(V_1 + H_1), \qquad \theta = \frac{1}{1 + (1+\phi)^{-\sigma}(1+r)^{\sigma-1}}.$$

Use the baseline parameters

$$V_1 = 20, \quad Y_1^d = 250, \quad Y_2^d = 320, \quad r = 0.03, \quad \phi = 0.03, \quad \sigma = 0.8.$$

1. Implement functions for human wealth, θ and the two-period allocation.
2. Compute C_1, C_2, total wealth and period-1 saving. Is the household saving or borrowing?
3. Increase the real interest rate from 3 percent to 7 percent. Decompose the change in C_1 into a component holding wealth fixed and a remaining wealth effect.
4. Plot C_1 as a function of $r \in [0, 0.10]$ for $\sigma = 0.5$, 1.0 and 1.5. Interpret how the intertemporal elasticity of substitution changes the result.
5. **Numerical cross-check.** Verify the closed-form C_1 for the baseline by maximising lifetime utility numerically. Build a fine grid of candidate C_1 values, compute C_2 from the IBC for each, evaluate

$$U(C_1) = u(C_1) + \frac{1}{1+\phi}u(C_2), \qquad u(C) = \frac{C^{1-1/\sigma}}{1-1/\sigma},$$

   and locate the maximiser. Report the closed-form C_1, the grid-search C_1, and their absolute difference. The two should agree to within 0.05. Briefly say what would have to be wrong in your code if they did not.

Part 3. Cash-flow exposure and borrowing constraints

This part connects the model to the lecture's Danish mortgage example. Compute monthly annuity mortgage payments using debt D, annual rate i and maturity M years.

1. For $D = 2{,}000{,}000$ DKK, $M = 25$ years, and a rate increase from 1 percent to 5 percent, compute the monthly and annual payment increase.
2. Assume the payment increase lasts for 3 years. Compute the present value of this cash-flow loss at a real rate of 3 percent.

3. Compare the implied annual consumption cut for two households:
   - a PIH household with a 30-year horizon;
   - a constrained household with MPC equal to 0.9.
4. Create a group-specific household profile by changing at least two parameters (debt, maturity, initial rate, new rate, horizon, or constrained MPC). Plot the PIH response and the constrained response for your profile, and explain why your chosen profile is plausible.
5. **Break-even constrained MPC.** For *your* group profile, find numerically the constrained MPC m^* at which the constrained household's annual consumption cut equals the PIH household's. Report m^* to three decimals and comment in two sentences on what the size of m^* implies about how easy or hard it is to identify constrained households empirically.

Part 4. Temporary versus permanent income shocks

Use the annuity factor

$$A(N, r) = \sum_{j=0}^{N-1} (1 + r)^{-j}, \qquad \theta_N = \frac{1}{A(N, r)}.$$

1. Plot θ_N for $N = 1, \ldots, 60$ with $r = 0.03$.
2. For a DKK 5,000 income increase, compare the consumption response when the increase is temporary and when it is permanent. Report the numbers for $N = 5$, $N = 30$ and $N = 60$.
3. Suppose 35 percent of households are constrained with MPC 0.9 and the rest follow the PIH. Compute the aggregate response to the temporary DKK 5,000 increase.

Part 5. Real data: consumption, income and saving

Use the provided CSV or fetch the following FRED series yourself: `DPIC96`, `PCEC96` and `PSAVERT`. The provided file is already quarterly.

1. Plot real disposable personal income and real personal consumption expenditure as indices with 2015Q1 = 100.
2. Plot APC $= PCE/DPI$ and the saving rate.
3. Compute quarterly log changes and estimate

$$\Delta \log C_t = \alpha + \beta \Delta \log Y_t^d + \varepsilon_t$$

   using NumPy. Report $\hat{\beta}$ and a 95 percent confidence interval.
4. Repeat the estimate excluding 2020Q1 to 2021Q4. Explain why this period may be special.
5. Make a scatter plot with the fitted regression line. Explain what the plot suggests and what it cannot prove.

Part 6. Your economic take-away

Write 250 to 400 words. What did you learn from this assignment? Mention

1. one result from the two-period model, citing the specific number you obtained;
2. one result from the mortgage cash-flow experiment, citing your own group profile;
3. one result from the data section, citing your own $\hat{\beta}$;
4. one limitation of your analysis.